

Lecture Notes on

Quantum Mechanics

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These notes cover quantum mechanics at the undergraduate level. The sources used are:

- Griffiths: *Introduction to Quantum Mechanics*

Potential sources to consult in the future:

- Sakurai: *Modern Quantum Mechanics*

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1 Fundamentals

1.1 The Stern-Gerlach Experiment

1.2 Bra-ket Notation

2 The Schrödinger Equation

2.1 Wave Functions

Classically, we work with masses and attempt to solve for their equations of motions $x(t)$ using corresponding Newton's laws or energy equations. The quantum mechanics approach is different; the idea is to instead solve for a particle's wave function $\Psi(x, t)$ and from there we can do anything. The goal then is to solve the Schrödinger equation

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V\Psi$$

After solving the Schrödinger equation for some initial wave function $\Psi(x, 0)$ the equation actually gives us all future values of the wave function $\Psi(x, t)$.

The statistical interpretation of the wave function goes as follows:

- Born's statistical interpretation of the wave function states that

$$\int_b^a |\Psi(x, t)|^2 dx$$

is the probability of finding the particle at $a \leq x \leq b$ at time t . Note that while Ψ is a complex valued function, $|\Psi|^2 = \Psi\Psi^*$ where Ψ^* is the complex conjugate of Ψ is always real and nonnegative.

- An obvious consequence is this is that even if you know everything about a particle (ie; you know its wavefunction), you still cannot predict with certainty the outcome of an experiment to measure its exact position.
- Before measurement, a particle's position is not well defined. Only after measurement can we say the particle has a "position," and by doing so we collapse the particle's wavefunction to a dirac delta centered around the measured position. This is known as the **Copenhagen Interpretation**
- The integral over all possible positions must be equal to one, since the particle must be located somewhere.

$$\int_{-\infty}^{\infty} |\Psi(x, t)|^2 dx = 1$$

If we have a wavefunction, we need to make sure its normalized for it to be physical. We can normalize it to obey the integral by multiplying it by a scale factor. A glance at the Schrödinger equation one can see that if $\Psi(x, t)$ is a solution, then $A\Psi(x, t)$ is also a solution for nonzero constant A .

- One can also show that the normalization integral does not change with time, so there is no need to constantly renormalized solutions as the particle evolves with time.

- If we interpret $|\Psi(x, t)|^2$ to be the probability, then we can say the expectation value of position is

$$\langle x \rangle = \int_{-\infty}^{\infty} x |\Psi(x, t)|^2 dx$$

Note that the interpretation of this quantity is the expected value of the position from measuring particles with wave function Ψ . Simply remeasuring the same particle over again will not work, since the first measurement will collapse the particle into some other wave function with exact position

- By differentiating the expected position with respect to time, one can obtain the expectation value of the velocity. It's customary for us to work using momentum so we have

$$\langle p \rangle = -i\hbar \int \left(\Psi^* \frac{\partial \Psi}{\partial x} \right) dx$$

- A more illustrative way to write the expectation values for position and momentum are:

$$\langle x \rangle = \int \Psi^* \cdot x \cdot \Psi dx$$

$$\langle p \rangle = -i\hbar \int \Psi^* \cdot \left(\frac{\partial}{\partial x} \right) \cdot \Psi dx$$

We say the the x operator "represents" position and the $-i\hbar \frac{\partial}{\partial x}$ operator "represents" momentum.

- All classical quantities can be represented as a function of position and angular momentum. Then the representation of a classical quantity $Q(x, p)$ can be written as

$$\langle Q(x, p) \rangle = \int \Psi^* Q \left(x, -i\hbar \frac{\partial}{\partial x} \right) \Psi dx.$$

- **Ehrenfest's Theorem** states that expectation values obey their classical laws. For example, the time derivative of momentum is

$$\frac{d\langle p \rangle}{dt} = \left\langle -\frac{\partial V}{\partial x} \right\rangle$$

2.2 Time Independent Schrödinger Equation

Let us assume that the potential $V(x)$ is independent of time. Then using separation of variables, we can substitute $\Psi(x, t) = \psi(x)\phi(t)$ into the Schrödinger equation giving us $i\hbar \frac{1}{\phi} \frac{d\phi}{dt} = -\frac{\hbar^2}{2m} \frac{1}{\psi} \frac{d^2\psi}{dx^2} + V$. Then for the equality to hold, both sides must be equal to some constant value E leaving us with two ordinary differential equations:

$$\frac{d\phi}{dt} = -\frac{iE}{\hbar} \phi$$

$$-\frac{\hbar^2}{2m} \frac{1}{\psi} \frac{d^2\psi}{dx^2} + V = E$$

The second equation is the **Time Independent Schrödinger Equation**. The first equation defines the characteristic time dependence and can be solved trivially $\phi(t) = e^{-iEt/\hbar}$. Separable solutions obey several important properties that we're interested in:

- These solutions are **stationary states**. What this means is while the wave function is obviously time dependent, $\Psi(x, t) = \psi(x)e^{-iEt/\hbar}$, the probability density is not:

$$|\Psi(x, t)|^2 = \Psi\Psi^* = \psi^* e^{+iEt/\hbar} \psi e^{-iEt/\hbar} = |\psi(x)|^2$$

We can further show that the expectation value of any classical quantity is also not time dependent in a stationary state:

$$\langle Q(x, p) \rangle = \int \psi^* Q\left(x, -i\hbar \frac{\partial}{\partial x}\right) \psi dx.$$

- The constant E represents the Hamiltonian of the system. Classically, the Hamiltonian is $H(x, p) = \frac{p^2}{2m} + V(x)$ so the corresponding Hamiltonian operator is $\hat{H} = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x)$. So the time independent Schrödinger equation can be rewritten as

$$\hat{H}\psi = E\psi$$

Then the expectation values of the Hamiltonian is

$$\langle H \rangle = \int \psi^* \hat{H} \psi dx = \int \psi^* E \psi dx = E$$

$$\langle H^2 \rangle = \int \psi^* \hat{H}^2 \psi dx = \int \psi^* E (\hat{H} \psi) dx = \int \psi^* E^2 \psi dx = E^2$$

This means the standard deviation is 0. So any ensemble of stationary states with the same wavefunction have their Hamiltonian measured to be the exact same E .

- Every solution to the Schrödinger equation can be written in the form of a sum of stationary solutions.

$$\Psi(x, t) = \sum_{n=1}^{\infty} c_n \psi_n(x) e^{-E_n t/\hbar}$$

So we can basically decompose any wave function into the sum of wavefunctions associated with specific energy levels. The quantity $|c_n|^2$ represents the probability that a measurement of the energy will return value E_n , so we also must demand

$$\sum_{n=1}^{\infty} |c_n|^2 = 1$$

$$\langle H \rangle = \sum_{n=1}^{\infty} |c_n|^2 E_n$$

2.3 Bound States

Consider the potential function given by

$$V(x) = \begin{cases} 0 & 0 \leq x \leq a \\ \infty & \text{otherwise} \end{cases}$$

Outside the potential well, $\psi(x)$ must be 0 because $V(x)$ goes to infinity. So we have $\psi(x) = 0$ for $x < 0$ and $x > a$. Inside the well, the Schrödinger equation gives us $-\frac{\hbar}{2m} \frac{d^2\psi}{dx^2} = E\psi$. The solution is just the simple harmonic oscillator with $k = \sqrt{2mE}/\hbar$. So the general solution is $\psi(x) = A\sin(kx) + B\cos(kx)$. Using the boundary condition that the wavefunction must go to 0 at $x = 0$ and $x = a$, we obtain

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right)$$

with corresponding energy

$$E_n = \frac{n^2\pi^2\hbar^2}{2ma^2}$$

Some properties of the ψ_n functions:

- ψ_n alternates parity around the center of the well. ψ_1 , the ground state, is even, ψ_2 , the first excited state is odd, ψ_3 is even and so on.
- Each ψ_n are orthonormal. Namely:

$$\int \psi_m(x) * \psi_n(x) dx = \delta_{mn}$$

- They form a complete basis for any function $f(x)$:

$$f(x) = \sum_{n=1}^{\infty} c_n \psi_n(x) = \sqrt{\frac{2}{a}} \sum_{n=1}^{\infty} c_n \sin\left(\frac{n\pi}{a}x\right)$$

This is simply a fourier series for $f(x)$, and this can represent any function $f(x)$ by **Dirichlet's Theorem**. To actually determine the coefficients, multiply both sides by ψ_m^* and then integrate

$$\begin{aligned} \int \psi_m(x)^* f(x) dx &= \sum_{n=1}^{\infty} c_n \int \psi_m(x)^* \psi_n(x) dx = \sum_{n=1}^{\infty} c_n \delta_{nm} = c_m \\ c_n &= \int \psi_n(x)^* f(x) dx \end{aligned}$$

Now consider the potential inside a harmonic oscillator $V(x) = \frac{1}{2}kx^2$. This potential is actually very illustrative because any general potential $V(x)$ can be taylor expanded into $V(x) \approx V(x_0) + V'(x_0)(x - x_0) + \frac{1}{2}V''(x_0)(x - x_0)^2 + \dots$ but notice that we can simplify by working with x_0 as an equilibrium point x_0 , so the $V'(x_0)$ term drops, and we can also choose to define V up to a scalar constant such that $V(x_0) = 0$. If we assume $x - x_0$ stays small, then we could describe any oscillatory motion with small amplitude.

The Schrödinger equation is now

$$-\frac{\hbar}{2m} \frac{d^2\psi}{dx^2} = \frac{1}{2}m\omega^2 x^2 \psi = E\psi$$

where $\omega = \sqrt{k/m}$. To solve this we introduce the **ladder operators** $\hat{a}_-$ and $\hat{a}_+$

- We define the operator $\hat{a}_{\pm}$ as

$$\hat{a}_{\pm} \equiv \frac{1}{\sqrt{2\hbar m\omega}}(\mp i\hat{p} + m\omega x)$$

The product is $\hat{a}_-\hat{a}_+ = \frac{1}{2\hbar m\omega} [\hat{p}^2 + (m\omega x)^2] - \frac{i}{2\hbar}(\hat{x}\hat{p} - \hat{p}\hat{x})$. Here, $(\hat{x}\hat{p} - \hat{p}\hat{x}) \equiv [\hat{x}, \hat{p}]$ where $[\hat{x}, \hat{p}] = i\hbar$ is what's known as the **commutator** of $\hat{x}$ and $\hat{p}$.

- Using the ladder operators, notice that $[\hat{a}_-, \hat{a}_+] = 1$ and we can rewrite the Hamiltonian and Schrödinger equation as

$$\hat{H} = \hbar\omega \left(\hat{a}_+\hat{a}_- + \frac{1}{2} \right)$$

$$\hbar\omega \left(\hat{a}_{\pm}\hat{a}_{\mp} \pm \frac{1}{2} \right) \psi = E\psi$$

- The $\hat{a}_+$ and $\hat{a}_-$ are accordingly named the **raising** and **lowering** operators. If ψ is a solution with energy E , then $\hat{a}_+\psi$ is also a solution with energy $E + \hbar\omega$ and $\hat{a}_-\psi$ is a solution with energy $E - \hbar\omega$. Notice that at the ground state, we are unable to lower the wave function any further, so we set $\hat{a}_-\psi_0 = 0$. Solving this differential equation gives us the ground state and energy:

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2}$$

$$E_0 = \frac{1}{2}\hbar\omega$$

- Then we can find the nth excited states by repeatedly applying the raising/lowering operators:

$$\hat{a}_+\psi_n = \sqrt{n+1}\psi_{n+1}$$

$$\hat{a}_-\psi_n = \sqrt{n}\psi_{n-1}$$

- It may also be helpful to represent the position and momentum operators using the ladder operators:

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a}_+ + \hat{a}_-)$$

$$\hat{p} = i\sqrt{\frac{\hbar m\omega}{2}}(\hat{a}_+ - \hat{a}_-)$$

You could also solve the Schrödinger's equation analytically using a power series, and obtain

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} \frac{1}{\sqrt{2^n n!}} H_n(\xi) e^{-\xi^2/2}$$

where $\xi = \sqrt{\frac{m\omega}{\hbar}}x$ and H_n are the **Hermite Polynomials**.

2.4 Scattering States

In the examples above, we have the total energy is less than the potential energy at the boundaries, so solutions are bound states. However, we can also have states where the particle enters from infinity into a finite potential, and then leaves to infinity again. Solutions to these cases are called scattering states.

A particle in free space isn't subject to any forces so $V(x) = 0$. This differential equation is much easier to solve, with stationary solutions

$$\Psi_k(x, t) = Ae^{i(kx - \frac{\hbar k^2}{2m}t)}$$

where $k \equiv \pm \frac{\sqrt{2mE}}{\hbar}$. A positive k means the wave travelling to the right, while a negative one is traveling to the left. The wave travels with phase velocity

$$v_{\text{quantum}} = \frac{\hbar|k|}{2m} = \sqrt{\frac{E}{2m}} = \frac{1}{2}v_{\text{classical}}$$

It's important to distinguish the group velocity from the phase velocity. The group velocity is the speed at which the "envelope" of superpositions travels at which represents the motion of the particle. Here, $v_{\text{group}} = \frac{d\omega}{dk} = v_{\text{classical}}$ so this problem is resolved. However, another problem is that this wave function is not even normalizable!

- To resolve this issue, instead of summing over separable solutions, we integrate over the variable k :

$$\Psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \Psi(k) e^{i(kx - \frac{\hbar k^2}{2m}t)} dk$$

- We can use **Plancherel's theorem** to determine $\psi(k)$ from the initial conditions:

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \Psi(x, 0) e^{-ikx} dx$$

2.5 The Hydrogen Atom

In 3 dimensions, Hamiltonian operator is now $\frac{1}{2m}(p_x^2 + p_y^2 + p_z^2) + V$. Notice that the momentum operator now becomes the gradient $\mathbf{p} = -i\hbar\nabla$. Thus:

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \Psi + V\Psi$$

However, most of the time the potential we will work with is spherically symmetric, in that it depends only on distance from the radius $V(r)$. Here it makes most sense to use spherical coordinates. By separation of variables, let

$$\psi(r, \theta, \phi) = R(r)Y(\theta, \phi)$$

The angular equation gives us

$$\frac{1}{Y} \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial Y}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2 Y}{\partial \phi^2} \right] = -\ell(\ell + 1)$$

Performing separation of variables again with $Y(\theta, \phi) = \Theta(\theta)\Phi(\phi)$ we now have

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d}{d\theta} \right) + \ell(\ell + 1) \sin^2 \theta = m^2$$

$$\frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = -m^2$$

The ϕ equation is easy to solve. The solution is just $\Phi(\phi) = e^{im\phi}$. Note that we also demand $\Phi(\phi + 2\pi) = \Phi(\phi)$ which implies m must be integer. Meanwhile, the solution to the θ equation is

$$\Theta(\theta) = AP_\ell^m(\cos(\theta))$$

where P_ℓ^m are the **Legendre Functions** defined as:

$$P_\ell^m(x) \equiv (-1)^m (1 - x^2)^{m/2} \frac{d^m}{dx^m} P_\ell(x)$$

and P_ℓ are the **Legendre Polynomials**. Thus, the solution to the angular equation are the **Spherical Harmonics**:

$$Y_\ell^m(\theta, \phi) = \sqrt{\frac{2\ell + 1}{4\pi} \frac{(\ell - m)!}{(\ell + m)!}} e^{im\phi} P_\ell^m(\cos \theta).$$

To solve the radial equation, instead we have $V(r) = -\frac{e^2}{4\pi\epsilon_0} \frac{1}{r}$. Solving this differential equation is genuinely so terrible so we'll skip that. The solution is

$$\psi_{nlm} = \sqrt{\left(\frac{2}{na}\right)^3 \frac{(n-l-1)!}{2n[(n+l)!]^3}} e^{-r/na} \left(\frac{2r}{na}\right)^l L_{n-l-1}^{2l+1}\left(\frac{2r}{na}\right) Y_l^m(\theta, \phi)$$

where $L_{q-p}^p(x) \equiv (-1)^p \left(\frac{d}{dx}\right)^p L_q(x)$ is the associated **Laguerre polynomial**. Here, $L_q(x) \equiv e^x \left(\frac{d}{dx}\right)^q (e^{-x} x^q)$ is the q th Laguerre polynomial. It's quite nasty, but generally it's extremely rare for the Schrödinger equation to even have a closed form solution like this. Generally speaking, we won't have this luxury with systems like the helium atom.

The allowed energies are

$$E_n = - \left[\frac{m}{2\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0} \right) \frac{1}{n^2} \right]$$

with the lowest possible energy state and

$$a \equiv \frac{4\pi\epsilon_0 \hbar^2}{me^2}$$

is the **Bohr Radius**, the smallest possible orbit.

3 Symmetries and Conservation Laws